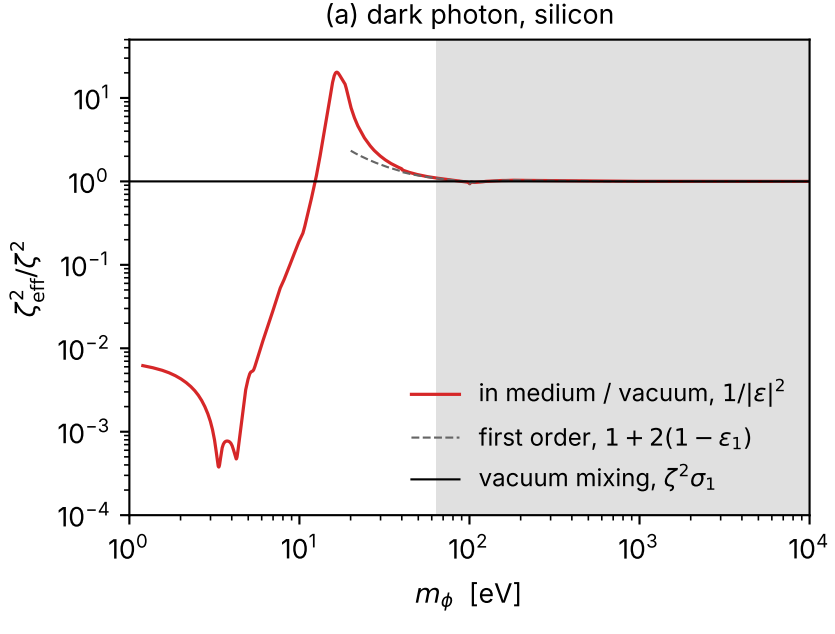
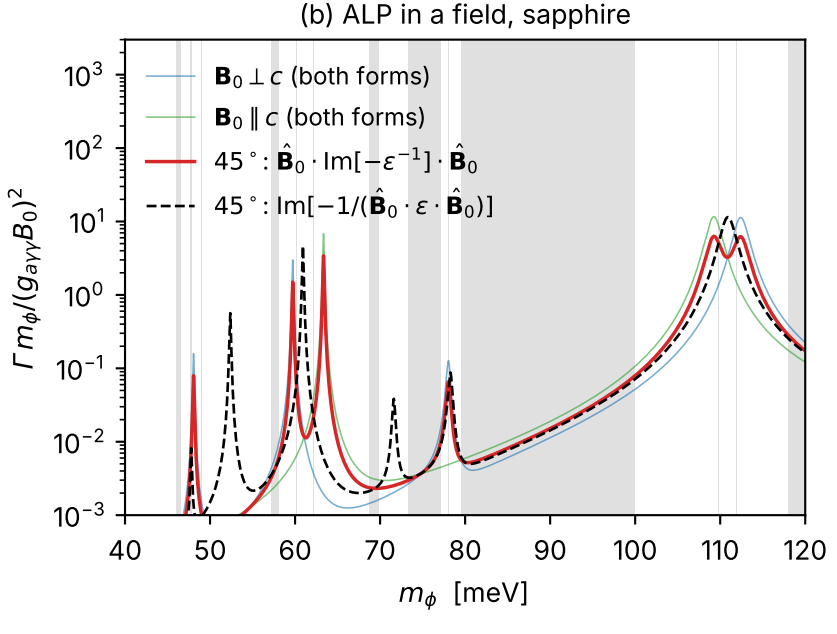

(a) dark photon, silicon
$\zeta^2_{\rm eff}/\zeta^2$
m_ϕ [eV]
in medium / vacuum, $1/|\varepsilon|^2$
first order, $1+2(1-\varepsilon_1)$
vacuum mixing, $\zeta^2\sigma_1$


(b) ALP in a field, sapphire
$\Gamma\, m_\phi/(g_{a\gamma\gamma}B_0)^2$
m_ϕ [meV]
$\mathbf{B}_0 \perp c$ (both forms)
$\mathbf{B}_0 \parallel c$ (both forms)
$45^\circ: \hat{\mathbf{B}}_0 \cdot \mathrm{Im}[-\varepsilon^{-1}] \cdot \hat{\mathbf{B}}_0$
$45^\circ: \mathrm{Im}[-1/(\hat{\mathbf{B}}_0 \cdot \varepsilon \cdot \hat{\mathbf{B}}_0)]$